

1 SH-004 — chain and phase gear

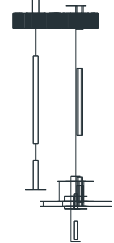
물리 검증·안전 인증·통전 승인이 아니다. E-stop, lid/service interlock, branch fuse, 독립 thermal fuse를 정상 firmware와 독립 구현하고 exact received component 정격·배선·보호소자를 실측 확인하기 전 통전하지 않는다.

전체 배치: GGM-FULL-ASM / 구동계: GGM R2. 기본 조립 투상도는 통합 전 참조이며 최신 전체 배치가 아니다. Revision: final-design-fabrication-closure-v0.8 · 상태: DIGITAL_DOCUMENT / PHYSICAL_NOT_RUN / USER_APPROVAL_REQUIRED

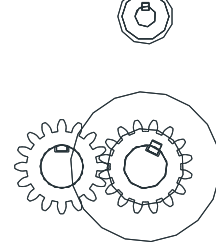
SH-004 chain phase gear

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 9

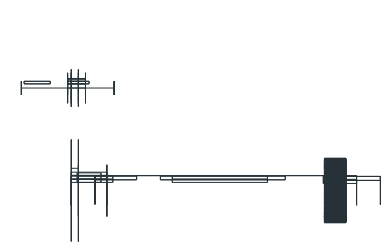
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 118.02 · Y 302.00 · Z 145.05 mm · 치수/공차는 개별 제작도면과 interface catalog 우선

Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

1.1 제작·검사 기준

- 단위: mm · 제3각법 · 축척: NTS(기입 치수 우선)
- 재료/구성: direct-keyed steel 35 sprockets / S45C phase gears / 35 chain
- 핵심 치수/공차: GGM_SH_12T direct-keyed to 12 h6 jackshaft with 4x4 key; GGM_SH_30T direct-keyed to CUT-05R 25 h6 with 6x6 key; keys carry torque; received maker retention hardware is axial-only and torque remains HOLD until receipt; sprocket radial TIR ≤ 0.10 mm; total axial shift+U95 ≤ 0.20 mm; chain alignment $\leq 0.20/150$ mm; midspan slack 2–3%; DRV-02 superseded; phase stack $M4 \times 4 = 3$ N·m
- 일반공차: ISO 2768-m. 개별 부품은 exports/final/manufacturing 도면과 interface_catalog.csv를 함께 검사한다.
- Source: 93ec6a9c0c2afa3ad72288a967f07b7c8d03c99e · 물리 검증: NOT_RUN